



Plan and Code

Plan it, write it, run it, change it

Name: _____

Date: _____

★ I can plan and write a code to reach a goal, then change it for a new goal.

You are the coder

Now you plan the whole code. Fill the empty move blocks so Bit reaches the goal ★. Plan it, write the blocks, predict the landing, then run it to check. Forward adds; back takes away.

Bit's number path

0	1	2	3	4	5 ★
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Goal 1 — start at 1, reach 5

start at 1

Fill the two move blocks to reach 5.

1 Plan and run

Fill the 2 empty blocks so Bit goes from 1 to 5. PREDICT the landing, then run to check. Write the blocks you used.

Goal 2 — start at 2, reach 3 (challenge)

start at 2

Use ONE forward block and ONE back block.

2 Challenge

Fill the 2 empty blocks so Bit goes from 2 to 3, using one forward block and one back block. Run it to check. Write what you used.

3 You try — change your code

Take your Goal 1 code (it reaches 5). Change ONE block so Bit lands on 4 instead. What did you change, and where does Bit land now?

Big idea

A coder plans the steps, writes the code, runs it to test, and changes it when the goal changes. You can read code, write code, fix code, and change code — you are a coder!



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Quick facilitation notes & answer key

What this worksheet teaches

The unit finale: students plan and write a full set of instructions to a goal, meet a harder goal that mixes forward and back, then change their instructions for a new goal. Read, write, fix, and change code — all in one page. No coding experience needed.

Before you start (no computer needed)

Paper and pencil only. Optional: a floor number line 0 to 5 with a star on 5, plus move cards, so students can lay out and test their plan before writing it.

How to lead it (about 20–25 minutes)

1. Show them first (you do). Model Goal 1 — count from 1 to 5 (4 steps forward) and split it: "forward 2, forward 2." Run it to land on 5.
2. Do one together (we do). Exercise 1: students plan their own blocks to reach 5, predict, then run. Exercise 2 (challenge): from 2 to 3 using one forward and one back (e.g. forward 2, back 1).
3. Let them try alone (you do). Exercise 3: change one block of the Goal 1 code so Bit reaches 4 (e.g. last forward 2 → forward 1).
4. Celebrate: "You can read, write, fix, and change code — that is what coders do."

Answer key (these are the verified correct answers)

Exercise 1 — start 1 → 5: any two forward blocks that total 4, e.g. forward 2, forward 2 (or forward 1, forward 3).

Exercise 2 — start 2 → 3 with one forward + one back: e.g. forward 2, back 1 (2 → 4 → 3). The pair must net +1.

Exercise 3 — change the last forward 2 to forward 1: 1 → 3 → 4. Bit lands on 4, one less, because the last step is now one smaller.

Why it works: the start sets the number and the moves run in order; plan the total change (forward adds, back subtracts) to match the distance to the goal.

If students get stuck — and ways to adjust

Watch for (Ex 2): using two forward or two back blocks. Re-read "one forward AND one back" and have them net +1 from 2 to 3.

Make it easier: allow a single move block for Goal 1 (forward 4).

Make it harder: ask for a second, different answer for Goal 1, or a plan that reaches 5 from start 0.

Ontario curriculum link (official wording)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

In plain terms: students follow (or write) step-by-step instructions, in order, to get a result.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

In plain terms: students read instructions someone wrote, change one, and explain what the change did.

You'll know it worked when

The student writes working instructions for each goal and correctly changes the Goal 1 code to land on 4.